

Day 3

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

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BIOS/POST Error Message	Hardware/Configuration Issue It Usually Points To
CMOS Checksum Error	Weak/dead CMOS battery or incorrect BIOS settings.
Keyboard Not Detected	Disconnected/faulty keyboard or damaged USB/keyboard port.
No Boot Device Found	Faulty/disconnected HDD/SSD or incorrect boot order.
Memory/RAM Error	Loose, faulty, or incompatible RAM.
CPU Fan Error	Disconnected, blocked, or failed CPU cooling fan.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

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Analysis and Meaning of the Beep Code

In systems utilizing an AMI (American Megatrends Inc.) BIOS, a POST beep code of 3 short beeps indicates a Base 64KB RAM Failure.

During the initial Power-On Self-Test, the BIOS checks the first 64KB block of system memory required to initialize fundamental hardware routines. A failure in reading or writing to this low-memory space prevents the PC from proceeding with the boot sequence or sending a signal to the display, resulting in a black screen and a continuous beep loop.

Troubleshooting Steps

Step 1: Reseat and Clean the RAM Modules

- **Procedure:**
 1. Complete a full system shutdown, turn off the Power Supply Unit (PSU) toggle switch, and disconnect the main power cable.

2. Open the chassis and unlatch the DIMM retention clips on the motherboard to remove all RAM sticks.
3. Clean the gold edge contacts of each module using a clean, soft microfiber cloth or an eraser (ensuring all rubber residue is removed).
4. Firmly push the RAM modules back into the memory slots until the retention clips automatically click into place.

Step 2: Perform Single-Stick Isolation Testing

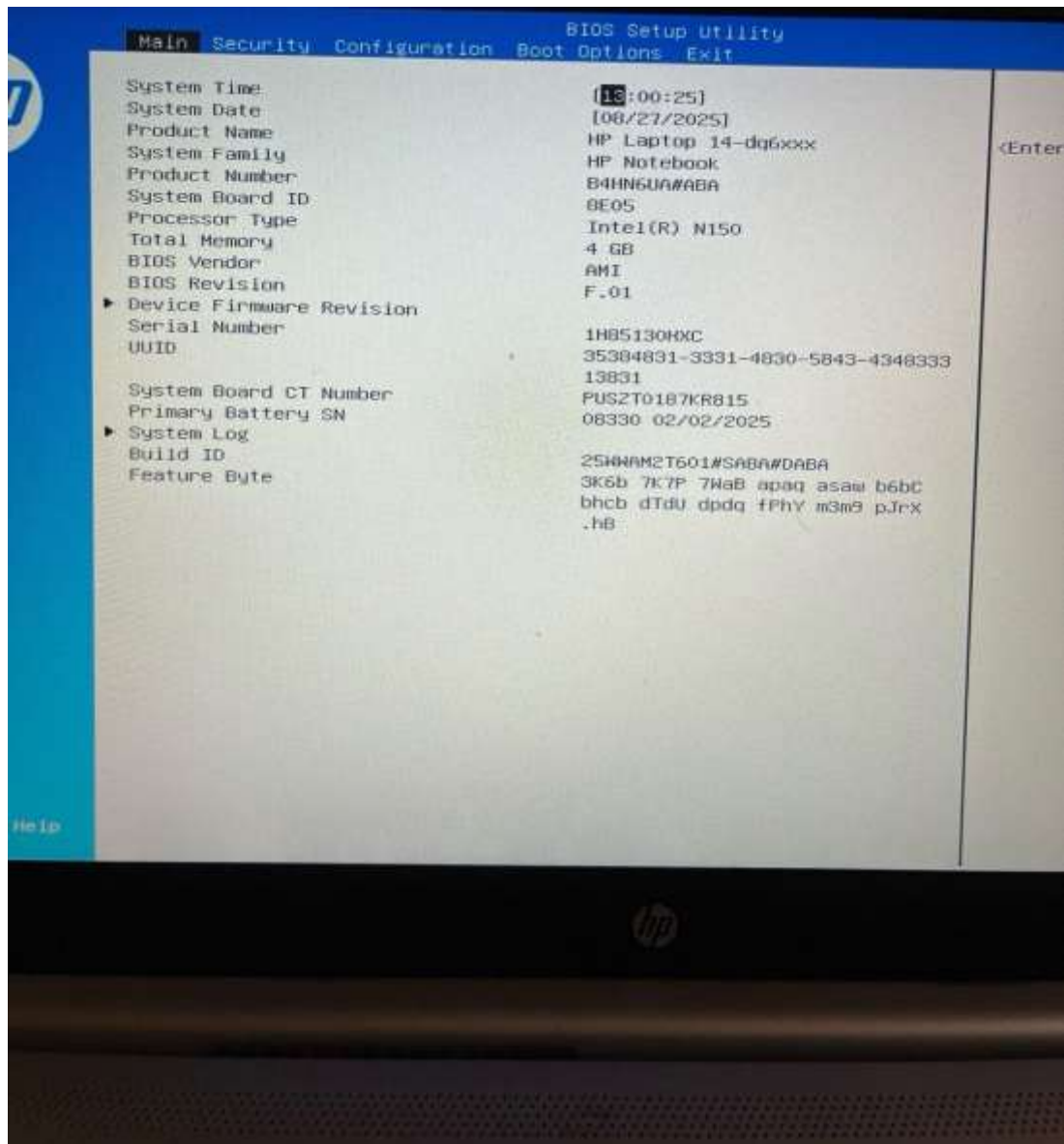
- **Procedure:**

1. Remove all memory modules from the motherboard.
 2. Insert only one module into the primary designated DIMM slot (typically slot A2 or DIMM 1 as per the motherboard manual).
 3. Power on the PC. If the system posts successfully, that module and slot are functional.
 4. Repeat this process with each individual stick of RAM across different slots to isolate whether a specific RAM module is defective or if a slot on the motherboard has failed.
- 3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.**

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Steps to Access BIOS Setup Utility

1. Shut down the HP laptop.
2. Press the Power button to start the laptop.
3. Immediately press the Esc key repeatedly to open the HP Startup Menu.
4. Press F10 to open BIOS Setup Utility.
5. On the main BIOS screen, locate the System Manufacturer and BIOS Version/Revision.
6. Record the manufacturer and BIOS version shown on the screen.
7. Exit the BIOS without making any unnecessary changes.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Hint: think about the CMOS battery and BIOS settings reset.

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Explanation of the Error Message

The error message "**CMOS checksum error – Defaults loaded**" occurs during the Power-On Self-Test (POST) phase of computer initialization.

Press to enter setup. <F12> Display Boot Menu.

BIOS Version: P01-A3
Release Date: 02/18/2011
CPU : Intel(R) Core(TM) i5-2500 CPU @ 3.30GHz
Stepping: 7 CPUID: 660206a7
Cache L1 Size: 256 KB
Cache L2 Size: 1024 KB
Cache L3 Size: 6144 KB
The IMC is operating with DDR3 1333
Dual Channel Mode
8192MB OK

USB Device(s) : 0 Keyboard, 2 Mice, 5 WSS, 2 HUBs
USB WSS #0:Generic-Compact Flash 1.00
USB WSS #1:Generic-SM/xD-Picture 1.00
USB WSS #2:Generic-SD/MMC 1.00
USB WSS #3:Generic-MS/MS-Pro/HG 1.00
USB WSS #4:Generic-SD/MMC/MS/MSPRO 1.00

Detected ATA/ATAPI Devices...
SATA Port0: WDC WD10EARS-0020B0
SATA Port4: ATAPI DVD A DH16A8SH
Error: CMOS checksum error - Defaults loaded
Press F1 to Continue. Press Del to Enter BIOS Setup.

What the Error Means:

During boot, the BIOS calculates a new checksum value from the data stored in CMOS and compares it with the saved value. When these two values do not match, it indicates that the stored BIOS data has been corrupted or lost. To prevent hardware malfunction, the system automatically overwrites the corrupted settings with factory safe-mode values ("Defaults Loaded").

Step-by-Step Resolution Process

1. Reconfigure BIOS Settings and Test Boot

- a. Turn on the PC and press the indicated key (Del, F2, or F10) to enter the BIOS setup screen.
- b. Reset the System Date and Time to the current values.
- c. Reconfigure custom boot preferences (e.g., Boot Drive Priority, UEFI vs. Legacy mode, or XMP profiles).
- d. Save changes and exit (F10).
- e. Test: Turn off the power supply completely for 5 minutes, then boot back up. If the error reappears, proceed to Step 2.



2. Replace the CMOS Battery (CR2032)

- Shut down the system, turn off the PSU switch, and disconnect the power cord.
- Open the PC chassis to expose the motherboard.
- Locate the round 3V coin-cell CR2032 battery on the motherboard.
- Gently press the small metal retaining clip outward to release the dead battery.
- Insert a fresh CR2032 3V Lithium battery with the positive (+) side facing upward until it clicks into position.

